Problem statement

JDK collections and streams cost heap twice: in what they hold and in what they drop.

- Baseline: boxes and nodes make 1 M `int`s take 20 to 48 MB.
- Volatile: 160 to 224 B of pipeline garbage on every stream call.
- Neither shows in the code, and escape analysis does not remove the stream’s.
- Goal: smaller footprint and less garbage, so a spike finds free heap.

References

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Package `java.util.stream` (Java SE 25 API)
- The `jmap` command (Java SE 25)
- Java Object Layout (JOL), OpenJDK Code Tools